

MAXWELL AFB, AL, USA

WMO: 722265

Lat: 32.383N

Lon: 86.350W

Elev: 171

StdP: 14.61

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	28.8	9.9	9.2	32.7	14.4	11.5	35.7	21.0	52.4	19.2	51.2	5.2	340	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	97.2	75.8	95.2	76.2	93.2	76.1	80.1	90.6	79.2	89.6	78.3	88.1	6.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	143.5	83.7	76.7	140.2	83.0	75.8	135.6	82.2	43.6	90.8	42.7	89.4	41.8	88.3	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.7	12.9	19.6	98.8	4.6	2.6	16.3	100.7	13.6	102.2	11.0	103.7	7.7	105.6
			17.8	81.9	4.7	1.3	14.4	82.8	11.6	83.5	9.0	84.2	5.6	85.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	47.7	52.9	58.8	65.8	74.2	80.8	82.4	82.5	78.2	67.7	56.2	51.6
	DBStd	14.23	10.33	9.38	8.60	6.52	5.29	3.70	2.87	3.36	4.72	7.50	7.84	9.19
	HDD50	391	169	72	26	2	0	0	0	0	0	28	94	
	HDD65	1951	540	350	225	69	6	0	0	0	59	279	423	
	CDD50	6466	97	154	299	477	750	923	1005	1007	845	549	215	145
	CDD65	2552	4	11	34	94	291	473	540	542	395	143	16	9
	CDH74	24950	16	57	278	840	2709	4913	5602	5586	3590	1220	98	41
	CDH80	11130	0	4	40	194	1084	2410	2657	2719	1637	374	10	1
Wind	WSAvg	5.4	6.2	6.5	6.6	6.1	5.2	4.6	4.2	4.4	4.7	5.2	5.2	5.7
Precipitation	PrecAvg	50.0	4.5	4.8	5.8	4.4	3.3	3.9	5.0	3.9	3.4	2.5	3.4	5.0
	PrecMax	69.5	9.8	12.8	13.0	10.7	9.3	9.7	9.6	10.1	9.3	8.2	9.3	13.7
	PrecMin	34.3	1.5	1.4	1.2	1.5	0.4	1.7	2.3	1.5	0.7	0.0	0.7	1.7
	PrecStd	7.8	1.9	2.3	3.0	2.2	2.2	1.8	2.0	1.7	2.2	1.8	2.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	80.0	84.1	87.1	93.9	99.0	98.7	101.2	97.4	90.5	81.0	78.6
		MCWB	65.4	67.8	66.4	69.2	72.7	75.5	77.2	75.4	74.0	72.3	68.1	68.1
	2%	DB	72.1	75.3	80.6	84.8	91.1	96.3	95.6	97.1	94.8	87.4	76.8	74.5
		MCWB	63.5	65.7	65.6	68.3	72.3	75.6	77.6	75.7	74.2	71.4	65.3	66.0
	5%	DB	68.1	72.0	77.1	82.2	88.8	93.8	93.6	94.6	92.0	84.3	73.3	70.7
		MCWB	61.9	62.7	63.4	67.6	71.7	75.8	77.4	76.9	74.1	69.6	63.7	65.3
	10%	DB	64.1	68.2	73.4	79.4	86.3	91.4	91.8	92.0	89.2	81.2	70.2	66.7
		MCWB	58.9	60.6	61.7	65.9	70.8	75.4	77.1	76.8	73.9	68.4	61.7	61.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.4	69.9	70.1	74.0	77.0	80.4	81.5	81.3	78.5	76.3	71.9	71.3
		MCDB	71.9	77.0	77.2	82.3	85.9	90.9	91.7	91.7	88.6	83.2	75.4	75.4
	2%	WB	66.0	67.4	68.3	71.9	75.1	79.0	80.2	80.1	77.4	74.8	69.2	68.6
		MCDB	70.1	73.1	75.6	79.8	85.5	89.9	90.5	90.8	87.1	82.0	73.8	72.0
	5%	WB	63.4	65.5	66.4	70.2	74.0	78.0	79.2	79.2	76.6	73.4	66.5	66.1
		MCDB	66.4	70.7	73.9	77.5	83.9	88.7	89.3	89.6	85.7	80.4	71.0	69.6
	10%	WB	60.4	63.0	64.4	68.5	73.0	77.2	78.3	78.2	75.7	71.7	63.7	62.9
		MCDB	64.0	66.6	71.2	75.7	82.4	87.4	87.9	87.6	84.1	77.8	67.9	66.2
Mean Daily Temperature Range	5% DB	MDBR	20.4	21.0	21.9	22.3	21.6	19.9	18.4	18.9	19.9	21.9	22.2	19.0
		MCDBR	23.5	23.0	24.9	24.8	24.5	23.1	20.8	22.1	23.7	23.4	23.7	22.1
		MCWBR	15.2	14.0	12.2	10.8	8.5	6.9	6.5	6.3	6.9	9.1	13.7	15.1
	5% WB	MCDBR	18.4	19.2	20.1	19.6	19.5	19.7	18.7	18.9	18.6	18.4	19.2	19.4
	MCWBR	14.0	13.5	11.6	10.9	8.6	7.4	6.9	6.5	6.4	8.7	13.7	15.4	
Clear-Sky Solar Irradiance	taub	0.324	0.332	0.361	0.392	0.442	0.481	0.538	0.528	0.434	0.374	0.347	0.327	
	taud	2.518	2.501	2.425	2.331	2.236	2.179	2.027	2.051	2.299	2.445	2.462	2.523	
	Ebn at Noon	285	294	291	285	270	259	243	244	265	275	273	277	
	Edh at Noon	27	30	35	40	45	47	55	53	40	32	29	26	
All-Sky Solar Radiation	RadAvg	886	1067	1433	1792	1972	1956	1900	1762	1560	1325	1001	759	
	RadStd	62	128	130	128	140	147	131	102	132	164	95	83	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+290	+189

Nomenclature: See separate page